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PI Email:	john.minna@utsouthwestern.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	214 648-4900
Organization Name:	The University of Texas Southwestern Medical Center		
PI Address 1:	Hamon Center for Therapeutic Oncology Research		
PI Address 2:	6000 Harry Hines Boulevard, NB8.206		
PI Web Page:			
City:	Dallas	**State:**	TX
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Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
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COI Name (Institution):	Asaithamby, Aroumougame (The University of Texas Southwestern Medical Center) Chen, David (The University of Texas Southwestern Medical Center) Chen, Min (The University of Texas Southwestern Medical Center) Gazdar, Adi M.D. (The University of Texas Southwestern Medical Center) Girard, Luc Ph.D. (The University of Texas Southwestern Medical Center) Shay, Jerry Ph.D. (The University of Texas Southwestern Medical Center) Story, Michael Ph.D. (The University of Texas Southwestern Medical Center) Xie, Jin (The University of Texas Southwestern Medical Center) Xie, Yang Ph.D. (The University of Texas Southwestern Medical Center)		
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Task Description:	The University of Texas Southwestern Medical Center (UTSW) NSCOR focuses on the development of quantitative dose risk estimates following HZE particle irradiation for the development of key genetic, epigenetic, gene expression, and cellular functional changes in the multi-step pathogenesis of lung cancer in both new human bronchial epithelial cell (HBEC) and transgenic mouse models of lung cancer. The risk of developing these changes are being compared in these models to that of gamma-irradiation. These dose risk assessments are designed to measure events of both radiation-induced promotion (modification of proliferation kinetics of already-initiated cells) as well as radiation-induced initiation (mutational) events. They also measure effects on development of cancer stem-like cells (cancer initiating cells). 3D organotypic culture and animal models also allow measurement and risk estimation in both tissue surrogates and in vivo lung tissues. These individual risk estimates can then be combined into a model for overall risk of developing lung cancer from exposure to galactic cosmic radiation. To achieve these goals, UTSW has assembled a team of scientists who are leaders in the study and translation application of the molecular pathogenesis of lung cancer (Minna, Gazdar, Shay); in radiobiology and DNA repair (Chen, Story); a panel of expert Internal and External Advisors (from other NSCORs and lung cancer research experts); as well as expert consultants and collaborators. This team has developed a novel immortalized HBEC system that can be genetically manipulated, studied in 2D monolayer and 3D organotypic cultures. Using these HBECs we have completed many studies with HZE particle irradiation showing that further progression toward lung cancer can be detected and quantified and that specific mRNA expression profiles for HZE irradiation exist. In addition, we have found that transgenic mouse models of lung cancer have increased lung cancer risk particularly after fractionated high LET exposure allowing us to study the early onset and progression of lung cancer in the whole animal in real time. The NSCOR has 4 Projects: 1. HZE Particle Exposure and the Risk for Human Lung Carcinogenesis; 2. Mouse Models of Lung Cancer after HZE Particle Irradiation; 3. Organotypic (3D) Human Models of Genomic Instability for HZE Particle Induced Lung Carcinogenesis; 4. Integrating Biomarkers of Lung Cancer Pathogenesis for Risk Assessment after HZE Particle Irradiation. These projects are supported by 2 Cores: Administrative Core and the Database, Biostatistics and Bioinformatics Core. The HZE particle irradiation is done at Brookhaven National Lab, and the biostatistical analysis shows the experiments are powered to help achieve NASA mandated risk estimate confidence levels. Finally, since this group has substantial UTSW institutional commitments, holds a NCI Special Program of Research Excellence (SPORE) in Lung Cancer, and a Center for the NCI Early Detection Research Network (EDRN) program, providing great synergism and additional resources available for the successful completion of this proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Lung cancer is the leading cause of cancer death in both women and men in the USA and in the Western world. While lung cancer is caused mostly by smoking exposure, a significant number of lung cancers arise in life time never smokers and thus is a sizable health problem in the USA. Indeed lung cancer in never smokers is the 7th leading cause of world-wide cancer related deaths. Likewise, 50% of all new lung cancer cases occur in former smokers (quitting over 5 years before). In all of these scenarios a major underlying question has been the role of other environmental carcinogens especially environmental radiation. One source of this is radon gas in the environment including radon gas exposure in homes with alpha particles (which represent high-LET radiation). Thus, the interaction of radiation in the environment on Earth including from alpha particles in the genesis of lung cancer and quantitation of the effect of such radiation on lung epithelial cells is of major health importance. In addition, in the past decade in the USA, Europe, and Asia there has been a resurgent interest in radiographic screening for lung cancer, particularly using serial computed tomography (CT) scans. In fact, the USA NCI has completed a large (50,000 person) National Lung Cancer Screening Trial (NLST) that randomized persons to have CT scans or not for the early detection of lung cancer. This trial has found a significant survival benefit for persons receiving CT scan screening. This raises a variety of questions concerning the use of this approach because of its cost and substantial population based radiation exposure. Greater than 25% of persons screened (current or former smokers) will have some lesions requiring further CT scan follow-up, when biopsies are performed only 1% of the overall group will have histologic evidence of lung cancer. In addition, many of the persons will require several CT scans in follow-up to determine if the lesion is growing to warrant a biopsy. Thus, there is considerable interest in knowing the contribution of CT scan doses of gamma radiation to the pathogenesis of lung cancer as well as other environmental exposures. This is particularly true since this radiation is being administered to individuals (current and former smokers) who are at highest risk of having preneoplastic lesions leading to lung cancer. Radiation is a known carcinogenic influence but the molecular events in lung epithelial cells in response to radiation need to be determined. A key element of this is the impact of radiation on lung epithelial cells with preneoplastic lesions. Our NSCOR uses innovative technology of human bronchial epithelial cells (HBECs) developed by us (coming from 50 different individuals) which have also been engineered to have known oncogenic preneoplastic changes (such as expression of oncogenic K-RASv12, knockdown of expression of mutant p53, expression of oncogenic mutant EGFR, expression of c-myc). Our studies are providing quantitative data following high-LET and low dose gamma irradiation on these HBECs. The primary benefits from this work will be the development of quantitative risk assessment models for astronauts of high-LET radiation in space in order to reduce the uncertainties in prediction of risk of astronauts developing lung cancer as part of long-term missions in space. Thus, the results will also have major impact and benefit to life on Earth that includes new knowledge of the effects on human lung epithelial cells of irradiation in terms of quantitative genetic and epigenetic changes and gene expression changes following irradiation which are of potential importance as markers of radiation exposure that could occur through CT scan screening studies, and environmental exposures, accidentally or through terrorism. A part of this will be to understand the quantitative response of DNA repair pathways to radiation including specific DNA repair components such as DNA repair enzymes and signaling pathways in lung epithelial cells. We are also determining quantitative changes in biologic phenotypes of lung epithelial cells in response to radiation which include colony formation in liquid and soft agar (anchorage independent growth) and differentiation and invasion in 3 dimensional (3D) cultures, as well as growth in vivo as either xenografts (for human cells) or endogenous lung cancers (for mouse models of lung cancer). From all of the above information we will identify novel molecular biomarkers (mRNA, DNA, protein) for lung carcinogenesis. Finally, we are also learning about inter individual variation of the response of HBECs to radiation with respect to the various genetic, epigenetic, mRNA and protein expression, and signaling pathway changes. Our lung cancer NSCOR is addressing the following specific critical path roadmap and gap questions: "We are improving the understanding of space radiation in lung cancer initiation, promotion and progression spanning cell and molecular biology at the cell, tissue and system biology level that will lead to significant reduction in the uncertainties in

risk projection models” ; "We are investigating the increased risk from space radiation as a function of age of exposure, age, latency, gender, tissue, radiation quality and dose rate” ; "We are developing tissue specific risk models using human 2D and 3D culture as well as animal models for lung cancer” ; "We are investigating how aberrant DNA damage processing, genomic instability, epigenetic effects including methylation, persistent oxidative damage, and non-targeted effects contribute to irradiation-associated lung cancer using a variety of surrogate endpoints that are being validated”. In summary, we are conducting research to investigate how biological endpoint and systems biology approaches can be used to integrate research on the cellular and molecular as well as tissue level of radiation damage that will lead to significant reductions in the uncertainties in risk projection models of lung cancer. Thus, our assembled working team is synergistically contributing more to assessing cancer risk associated with space radiation than would be accomplished by study of the individual components in isolation.

Our lung cancer NSCOR is addressing the following specific critical path roadmap and gap questions:

• We are improving the understanding of space radiation in lung cancer initiation, promotion and progression spanning cell and molecular biology at the cell, tissue and system biology level that will lead to significant reduction in the uncertainties in risk projection models

• We are investigating the increased risk from space radiation as a function of age of exposure, age, latency, gender, tissue, radiation quality and dose rate

• We are developing tissue specific risk models using human 3D culture and animal models for lung cancer

• We are investigating how aberrant DNA damage processing, genomic instability, epigenetic effects including methylation, persistent oxidative damage, altered senescence, and non-targeted effects contribute to irradiation-associated lung cancer using a variety of surrogate endpoints that are being validated

Overall we are conducting research to investigate how biological endpoint and systems biology approaches can be used to integrate research on the cellular and molecular as well as tissue level of radiation damage that will lead to significant reductions in the uncertainties in risk projection models of lung cancer. As will be described in this report, we are confident and will provide compelling evidence that the sum of the efforts of our assembled team are significantly contributing more to assessing cancer risk associated with space radiation compared to the individual components.

Project 1. HZE Particle Exposure and the Risk for Human Lung Carcinogenesis (Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation). Michael Story, Ph.D. Lead, Adi F. Gazdar, M.D. Co-Lead, John D. Minna, M.D. Co-Lead.
Specific Aim 1. Identify the radiation parameters, (dose, dose rate, Z, energy, LET) that drive radiation-induced cellular transformation to malignancy in normal HBEC/HSAECs.

Specific Aim 2. Determine the role of gender, lung cell of origin, and inter-individual variation in HZE particle-induced lung carcinogenesis using a large cohort of normal HBEC/HSAEC lines derived from more than 50 individuals.

Specific Aim 3. Determine the nature and frequency of genetic and epigenetic changes, and pulmonary stem cell markers associated with HZE particle-induced lung cancer.

Specific Aim 1 and 3:

Radiation-induced cellular transformation (ability to grow in soft agarose) of normal human bronchial epithelial cells: In this report we describe the genomic analysis of representative transformed cell lines (acquiring the ability to grow as colonies in soft agarose) after g-, 56Fe, and 28Si (both at 1 GeV nominal energy). Thirty four transformed cell lines isolated from cells capable of anchorage independent growth in soft agarose have been collected. These 16 cell lines were chosen as representative of all clones based upon radiation exposure and the clone’s robustness in growing in soft agar, from similar to the parental cell line to 2,000-fold higher. While the molecular analysis of KRAS and p53 are ongoing, no mutations of p53 have been identified by sequencing of p53 mutational hotspots, 3 clones have been identified that are heterozygous for mutations in KRAS (V12). As one of those clones arose spontaneously, it is unlikely that such a mutation is radiation-induced. We have found that there is differential expression of transforming growth factor beta induced mRNA (BigH3 mRNA). There is only a slight separation of transformants based upon gene expression.

Chromosomal Copy Number changes in transformed HBECs: The transformants were compared to a reference human DNA SNP map using the Illumina OmniQuad 1M SNP chip. This chip interrogates one million single nucleotide polymorphisms (SNPs) across the human genome, and because the signal for a given locus can be quantified, chromosomal copy number variation (CNVs) can be determined. All clones contained the amplification of chromosomes 5 and 20, known to be present in the parental HBEC3 KT cell line however, there were a number of other differences in chromosomal copy number, particularly in Chromosome 7 (Ch7). There are very common regions of deletion and amplification between the various groups. Interestingly, the HZE irradiated samples display much larger and contiguous regions of amplification and deletion. These alterations appear to be single copy changes, and, when gene expression is overlaid there are no statistically significant differences in gene expression that parallel CNV. There are other regions of the genome where there is a corresponding change in gene expression with CNV. There is one of particular interest, the tumor suppressor gene FHIT, down-regulated an average of 6.5-fold, is in a 600 kb deleted region of the genome of transformed cells. There are seven clones where smaller regions of both copies of the gene are deleted. Those regions of the gene deleted in both copies are depicted by the deep red color. This intriguing result is undergoing follow-up.

Low dose rate exposures and cellular transformation: In the fall of 2011 we began laying the foundation for performing low dose and dose rate heavy particle exposures. We defined the operational parameters under which we perform these experiments during NSRL 11C and 12A. We irradiated cells with dose rates as low as 0.002 Gy/min up to doses of 0.25 and 0.50 Gy. These doses represent that maximal dose for transformation (0.25 Gy) and the dose at which the transformation frequency declined (0.50 Gy). We hypothesized that we would see an inverse dose rate effect for cellular transformation with the likelihood that at 0.50 Gy the transformation frequency will continue upward rather than decline. HBEC cell lines from three individuals will be challenged in this way. In addition to the standard two-dimensional cell culture methods previously employed, the same experiments challenged cells grown in 3-D cultures. Here the initial endpoint will be de-differentiation and re-entry into the cell cycle, the result being a morphologic change in cell culture appearance that can be seen by eye. We predicted that these experiments will provide the optimal dose and dose rate for

determination of cellular transformation frequency against which we can test all 50 of our individual cell lines in future studies.

Future Plans: We will continue to define the genetic and epigenetic changes associated with spontaneous as well as radiation-induced cellular transformation. We need to continue to study a larger panel of HBECs and HSAECs to determine inter individual variation in transformation after exposure to HZE particles. We will also broaden the genetic diversity of our HBEC panel. It is important to generate a quantitative relationship between cellular transformation and oncogenesis. As of this time, no transformed clone generated spontaneously or by radiation exposure of HBEC3 KT cells has been shown to be tumorigenic in immune deprived mice. We will test these transformed clones in more immune deprived gamma knockout mice. It is likely that the time post-irradiation has not been sufficiently long enough for the genomic instability described last year to result in sufficient genetic or epigenetic alterations that would lead to oncogenesis. Therefore, we are continuing to culture several of these transformed clones for several more months in order to test their oncogenic potential in the future based upon cellular and molecular changes identified with continued culture.

Project 2. Mouse Models of Lung Cancer after HZE Particle Irradiation. Jerry W. Shay, Ph.D. Project Lead.

Using transgenic mouse models of lung cancer this project will determine the role of stress/inflammatory cell responses in murine lung cancer arising after HZE particle exposure. Using genome wide technology, we are comparing the genetic changes in radiation induced versus spontaneous arising murine lung cancers. All of these results and unique bio reagents have been made available to other NASA Lung Cancer Consortium members.

Aim 1: To test the hypothesis that the stress/inflammatory signatures are biomarkers of protracted/fractionated high LET induced invasive lung cancer.

Aim 2: To test if single dose acute but not fractionated equivalent dose high LET IR leads to increased stem cell killing. We will determine if fractionated high LET IR but not single acute doses of IR "wake up" quiescent stem cells leading to cancer progression.

Aim 3: To determine the molecular events (using genome-wide technologies) occurring in murine HZE particle IR associated lung cancers compared to mouse lung cancer arising in the absence of HZE IR. We will use our current mouse model and compare to other mouse models of lung cancer as are developed by other NASA Lung Cancer Consortium members.

Molecular studies on lung cancer progression in humans have revealed that one of the hallmark changes includes oncogenic K-ras mutations. Expression of oncogenic K-ras in the murine lung initiates the formation of lesions that mimic lung cancer progression in humans and permits evaluation of lung cancer progression in vivo after exposure to varying types, doses and dose rates of irradiation. We have chosen to use the LA-1 K-ras lung cancer susceptible mouse model to examine the cancer promoting effects of heavy ions such as Si and Fe as well as solar particle events (SPE) simulations. The long term goal of the study is assess the risk to lethal lung cancers and to then extrapolate the data to human risk projections. A key NASA objective is to extrapolate from cell and animal models to human risk and our initial results suggest that this will be possible. The murine model (LA-1) randomly expresses mutated K-ras in a subset of lung cells resulting in initiation and formation of lesions which mimic lung cancer progression in humans. Greater than 50% of the mice with oncogenic K-ras expression die in less than a year with a small percent of mutant mice living to a maximum of about 600 days approximately 200-300 day less than wild-type mice. About~15% of LA-1 mice spontaneously develop invasive (lethal) non-small cell lung carcinomas after about one year.

In previous studies we sought to determine if X-ray or charged particle radiation exposure affected the initiation or progression of lung cancer in mouse models. We used whole body irradiation (as an acute 1.0 Gy dose or a fractionated dose of 0.2 Gy per day for 5 days) of an oncogenic K-ras mouse model (LA-1), which is predisposed to benign tumors. We observed that irradiation did not impact the number or size of benign lung tumors initiated in the animals (reported previously). However, approximately one year after irradiation, histopathological analysis of tumors from LA1 K-ras mutant mice irradiated with the fractionated 1.0 Gy dose of 56Fe particles (1GeV) demonstrated a dramatic progression in grade of lesion compared to all other groups. Importantly, LA1 K-ras mice that received fractionated doses of 56Fe particles (but not an acute dose) showed a significant increase in the number of invasive adenocarcinomas. During the last year we administrated whole body proton irradiation, 2.0 Gy over 2 hours with a wide range of energies (50 MeV-150MeV) simulating a solar particle event. Results from the solar particle event (SPE) simulation demonstrated survival differences between irradiated and control LA-1 mice. Further analysis of the irradiated LA-1 animals demonstrated a dramatic progression in tumor grade by histopathological analysis. While the number of initiated lesions between the irradiated and unirradiated mutants were similar, mice exposed to a 2.0 Gy protons (SPE simulation) also demonstrated a significant increase in occurrence of invasive adenocarcinomas. These advanced invasive lesions observed in animals exposed to a Fe and Proton IR closely resemble adenocarcinomas clinically observed in humans. Experiments in progress include a cohort of mice (n = 40) that were exposed to single dose protons at 50 MeV. Due to an increasing understanding of the role of the tumor microenvironment and inflammation on lung cancer progression, experiments were initiated to determine if signatures of advanced disease could be detected significantly earlier than one year after irradiation. Thus, we repeated the above experiments and isolated lung tissue 70 days after irradiation (a time point that where only low grade hyperplastic lesions are present and over 200 days prior to the histological observation of advanced invasive tumors). We then conducted genome wide expression analysis of whole lung (15 mice each of unirradiated, fractionated, and acute 56Fe particles irradiation) and analyzed the results by hierarchal clustering and principal component analysis (data shown in last year’s report). Based on these results, we then sought to extrapolate from experimental mouse data to human archival lung cancer expression datasets. After matching genes between mouse and human array platforms we then compared the signatures from our mouse data to determine if there was a correlation with overall survival in patients with lung cancer from several large data sets. Based on a small gene classifier set, we were able to show across several independent data sets that the space fractionated Fe mouse lung signatures had high prognostic value when applied to human lung tumors for determining the survival outcome of human patients with lung cancer. Using integration of biological networks and gene expression data along with other bioinformatics tools, we were able to identify distinct stress/inflammatory responses after irradiation but prior to histopathological detection of invasive lesions. Additional validation of the proteins and pathways involved are in progress. In summary, these results provide robust evidence that sustained exposure to space radiation in the presence of pre-malignant initiated cancer lesions could affect astronaut increase risk of fatal cancers in future long-term missions.

Future Plans: Carcinogenesis is an adaptive process between nascent tumor cells and their microenvironment, which includes modifying inflammatory responses from anti-tumorigenic to pro-tumorigenic. Radiation exposure can induce

stress/inflammatory responses that both inhibit and promote carcinogenesis. We have demonstrated that radiation exposure accelerates lung cancer progression in the K-ras LA-1 lung cancer mouse model with dose fractionation creating a more permissive environment for cancer progression. We have also found that SPE simulations increase lung cancer progression in this mouse model. Comparative genomic analyses revealed a disrupted inflammatory signature that develops in mouse lungs after HZE particle irradiation but before tumors develop ("fractionation signature") behind this progression that is capable of predicting survival of NSCLC patients after surgical resection. These results demonstrate that radiation exposure can cooperate with pre-initiated lesions by disrupting inflammatory pathways in mouse transgenic cancer models and that clinical relevant similarities exist in the disruption of inflammation during human lung carcinogenesis.

Significance: This is an important advance in elucidating mechanisms of cancer progression using transgenic mouse models of cancer susceptibility that have direct implications for human carcinogenesis and may lead to more accurate prediction models for increased risks of lung cancer. In the next review period, we will analyze the results of the single dose proton and SPE simulation to determine if similar stress/inflammatory signatures are present as we found with Fe IR. We will continue treating mice with low doses and dose rates of Fe and Si to increase the confidences of our results that protracted and low dose IR does increase carcinogenesis in transgenic mouse models of lung cancer. In addition, we will be analyzing gender and age effects on previous and ongoing experiments. As part of a separate NASA grant (NNX11AC15G) we are also examining the role of a potent oral available biological countermeasure and have preliminary data showing dramatic decreases in lung cancer progression when LA-1 K-ras mice are provided the countermeasure (CDDO-EA) in their chow ~2 to 3 days prior to irradiation without additional countermeasure treatments after IR. These experiments will be detailed in the annual report for this project

Project 3. Human Organotypic (3D) Genomic Instability Model for HZE Particle-Induced Lung Carcinogenesis. David Chen, Ph.D. Project Lead, Aroumougame Asaithamby, Ph.D. Co-Lead.

Overall Objective: With the ultimate objective of estimating cancer risks to humans from HZE particles in space flight, in this project we propose to understand the DNA damage responses of human lung epithelial cells to HZE particles, since the failure of these responses to correctly repair radiation damaged DNA and increased cancer predisposition is established. In both pre-cancerous lesions and cancers, activated oncogenes induce stalling and collapse of DNA replication forks, which in turn lead to the formation of DNA DSBs. This continuous formation of DNA DSBs may contribute to the genomic instability that characterizes the vast majority of human cancers. In what could be seen as an analogous process, the complex DSBs along the densely ionizing track induced by HZE particles are difficult to repair and, in some instances, are irreparable. These unrepaired or mis-repaired DNA lesions likely contribute to the higher relative biological effectiveness for cell killing, chromosomal aberrations, mutagenesis, and carcinogenesis seen in HZE particle-irradiated cells. While the DNA damage response machinery may play other important roles in tumorigenesis, there are insufficient data to determine the relationship between early biological effects of radiation, (e.g. unrepaired and/or mis-repaired DNA lesions), and the probability of cancer in humans. It is this missing knowledge that in part leads to large uncertainties in projecting cancer risks associated with occupational or environmental-space radiation exposures. As we are specifically interested in risks related to lung cancer development upon exposure to space radiation, we have developed two novel approaches. We have: 1) made human bronchial epithelial cells (HBECs) stably expressing EGFP-53BP1 which allows real time, and continuous monitoring of the occurrence of 53BP1 foci (a hallmark of DSBs) and their persistence over time (indicating lack of or mis-repair); and established a novel organotypic (3D) human lung epithelial cell culture model where HBEC and human small airway epithelial cells (HSAECs) grown in Matrigel overlying a lawn of lung stromal fibroblast cells differentiate into tissue-specific structures mimicking those found in normal lung. Using EGFP-53BP1 monitoring and the 3D organotypic model we have found that cells in 3D culture (compared to the same cells growing in standard "2D" monolayer tissue culture) exhibit larger numbers of unrepaired DNA lesions (EGFP-53BP1 foci) following 1 Gy 56Fe-ion irradiation (~35 vs. ~20% at 72 hr. after IR). Based on these findings, we hypothesize that unrepaired DNA lesions in HZE particle-irradiated tissues generate a "signal" that provides selection pressure for the abrogation of DNA repair and checkpoint mechanisms and changes leading to genome instability and increased rates of cell transformation (soft agar colony formation) and full-fledged tumorigenesis (in vivo in xenograft models). To test this hypothesis, we will utilize our novel human lung epithelial cell models to: a) Demonstrate that the HZE particle induced persistent DNA damage and genomic instability (by monitoring of 53BP1 foci) in human organotypic lung respiratory cell models is dependent on the dose, dose rate, energy, "Z" number and frequency (acute versus fractionated) of HZE particle radiation; b) Determine the mechanisms of HZE particles induced human lung carcinogenesis in the organotypic model. The data obtained from this aim will provide biological markers correlated to lung carcinogenesis for the development of risk assessment and radiation protection for astronauts during future space missions.

Specific Objectives:

Aim 1: Determine if HZE particle-induced genome instability in organotypic lung (HBEC/HSAEC) models is dependent on the dose, dose rate, energy, "Z" number and the frequency (acute versus fractionated) of HZE particle radiation.

Aim 2: Determine the mechanisms of HZE particle-induced human lung carcinogenesis in the organotypic HBEC/HSAEC model.

Sub Aim 2.1: To determine whether unrepaired DNA lesions generated by HZE particles constitutively activate DNA damage signaling.

Sub Aim 2.2: To determine the effects of HZE particles on the differentiation capacity of HBEC/HSAECs.

Sub Aim. 2.3: To determine the effects of HZE particles on proliferation, viability and replicative senescence of organotypic HBEC/HSAECs.

Sub Aim. 2.4: To test the role of the microenvironment on HZE particle-induced lung carcinogenesis.

Sub Aim 2.5: To determine the effect of HZE particles on the genetic or epigenetic alterations and transforming potential of organotypic HBEC/HSAEC cultures.

Summary of Current Progress: During this funding period, we have examined how cells in monolayer and 3D cultures respond to low dose rate iron particles. Using two different dose-rates, i.e., 0.2 and 20cGy per minute and two different doses, i.e., 25 and 50 cGy, of iron particles (1GeV/n) we found that the efficiency of clustered DSBs repair was independent of the dose-rate, but it was dependent on the total dose both in monolayer and 3D organotypic cell cultures. Similarly, both the extent of cell killing and the levels of gross-chromosomal aberrations were also not affected by the

Task Progress:

dose-rate in monolayer HBECs. Overall, our data suggest that the radiosensitivity, cellular ability to repair clustered DSBs, and the induction of gross-chromosomal aberrations were not influenced by the dose-rate of the Fe particles.

1. To determine whether HBECs survival efficiency was affected by the different dose-rates of the Fe particles. We irradiated exponentially growing monolayer cultures of HBECs-3KTs with Fe particles of 0.2 and 20 cGy per minutes and then examined the extent of cell survival by colony formation assay. These results suggest that dose-rate has no influence on the cell survival. However, it will be interesting to determine whether HZE particles other than Fe, exhibit similar dose-rate independent cell survival efficiency.

2. To determine whether HBECs grown in monolayer culture repair clustered DNA double-strand breaks induced by low-dose rate of Fe (1GeV/u) particles exposure. There is growing interest in quantification of the biological effects of low doses of ionizing radiation, particularly when they are delivered at low dose rate, a situation relevant to most environmental exposures, including the planned manned mission to Mars. As low dose-rate irradiation can have diverse effects on end points of potential relevance to human risk, examination of DSBs repair in the context of low-dose rate may reveal early indicators of carcinogenic process. To determine whether HBEC-3KTs cells grown in monolayer culture repair clustered DNA lesions induced by low-dose rate Fe particles, we irradiated monolayer culture of HBECs stably expressing EGFP-53BP1 with 0.5 Gy of Fe particles at 0.2 and 20 cGy/min dose rate. Following irradiation, samples were fixed with 4% paraformaldehyde at 30 min, 24 hr and 72 days. Subsequently, we quantitated the number of both induced as well as persistent EGFP-53BP1 foci at different post-IR times. Both 20 and 0.2 cGy per minute dose rates of Fe particles induced fewer number of EGFP-53BP1 foci per cell, however the number of EGFP-53BP1 foci numbers per cells in 0.2cGy per min irradiated samples were lower than that of 20 cGy per min samples. This could be due to the ongoing repair of initially induced DSBs during the long irradiation period. Nevertheless, the number of unrepaired DSBs was not significantly different between 20 and 0.2 cGy per min, indicating that the dose rate of Fe particles has no influence on the kinetics of DSB repair processes.

3. To determine whether HBECs grown in organotypic culture repair DNA double-strand breaks induced by low-dose rate of Fe (1GeV/u) particles exposure. To determine whether HBEC-3KTs cells grown in organotypic culture repair clustered DNA lesions induced low-dose rate Fe particles similar to monolayer cultures, we irradiated three-dimensional cultures of HBECs stably expressing EGFP-53BP1 with 0.25 and 0.5 Gy of Fe particles at 0.2 and 20 cGy/min dose rate. Following irradiation, samples were fixed with 4% paraformaldehyde at 30 min, 24 hr and 72 days. Subsequently, we quantitated the number of both induced as well as persistent EGFP-53BP1 foci at different post-IR times. The number of unrepaired DSBs was not significantly different between 20 and 0.2 cGy per min, indicating that similar to monolayer culture, the dose rate of Fe particles has no influence on the kinetics of DSB repair processes in 3D HBECs.

4. To determine whether the levels of gross-chromosomal aberrations in monolayer cultures of HBECs were modulated by the low-dose rates of Fe (1GeV/u) particles exposure. To examine the relationship between the gross-chromosomal aberrations and the dose-rate effects of Fe particles, we irradiated monolayer cultures of HBEC-3KT with 0.5 Gy of Fe particles at 0.2 and 20 cGy/min dose rate. At 16 hours following irradiation, colcemid was added and the metaphase spreads were prepared. Subsequently, chromosome aberrations were scored. Exposure of HBEC-3KTs with 0.5 Gy of Fe particles at 0.2 and 20 cGy per min induced only a few chromosome aberrations. Interestingly, the extent of chromosome aberrations was not significantly different between 20 and 0.2 cGy per min, indicating that the dose rate of Fe particles has no influence on the induction of total chromosome aberrations in monolayer cultures of HBEC-3KTs.

Future Plans:

1. To test the role of the microenvironment on HZE particle-induced lung carcinogenesis. Interactions between epithelial cells and fibroblasts influence proliferation and transformation of epithelial cells, while the abundant, non-neoplastic fibroblastic portion of the stromal compartment in most solid cancers (often referred to as “cancer associated fibroblasts”, CAFs) plays an active role in tumor proliferation, invasion and metastasis by orchestrating tumor promoting inflammation in an NFkB dependent manner. In response to DNA damage, signals from the microenvironment persist for long periods and thus may contribute to tumorigenesis. Our 3D organotypic culture system contains both HBEC/HSAECs and fibroblasts and the effects we have seen could come from HZE radiation of epithelial cells, fibroblasts or both. Thus, we need to establish whether HZE particle radiation influences fibroblasts to assume a “CAF” like phenotype and whether CAFs isolated from lung cancers can influence HZE particle associated lung carcinogenesis.

Experimental Approach:

I. To establish the role of HZE particle radiation on fibroblasts we will compare non-irradiated vs. Fe, Si, O and H+ irradiated normal lung fibroblasts (NLFs) and also human lung cancer associated fibroblasts (CAFs, available through our NCI Lung Cancer SPORE, P50CA70907) on the proliferation, differentiation and number and persistence of EGFP-53BP1 foci in HBECs/HSAECs in 3D organotypic Matrigel culture.

II. Likewise, since cancer cells are able to generate CAFs, we will test whether Fe, Si, O, and H+ irradiated HBECs/HSAECs when subsequently mixed with NLFs can alter their physiologic profiles and demonstrate enhanced migration, loss of Caveolin-1 expression, and transdifferentiation to express epithelial cytokeratins

2. To verify whether HBECs grown in organotypic culture repair DNA double-strand breaks induced by fractionated Fe (1GeV/u) particles exposure. To verify whether cells grown in organotypic culture repair clustered DNA lesions induced fractionated doses Fe particles (0.2 Gy per day for five consecutive days), we will irradiate monolayer and three-dimensional cultures of HBECs stably expressing EGFP-53BP1 with fractionated doses of Fe particles. Samples will be fixed with 4% paraformaldehyde at 30 min, 24 hr and 5 days after irradiation. Subsequently, number of induced and persistent EGFP-53BP1 foci will be quantified in these samples.

3. To determine whether unrepaired DNA lesions generated by HZE particles constitutively activate DNA damage signaling. Activation of DNA-damage signaling protein kinases (ATM, ATR, DNA-PK, Chk1, Chk2) and phosphorylation of their substrates (for example, p53, 53BP1, SMC1, and KAP1) are essential for the implementation of cell cycle checkpoints or induction of apoptosis upon irradiation. These proteins can serve as early biomarkers or end points for carcinogenesis as the loss of these kinases or their substrates leads to cancer predisposition. It has been shown that persistent DNA damage signaling emanating from unrepaired DNA lesions creates selection pressure for mutations in apoptosis/senescence genes, thereby, abrogating barriers to cancer progression. Experimental Approach: We will irradiate organotypic HBECs cultures with 1 Gy of Fe particles (at 1000 MeV/n), and periodically (30 min to 30 days) assess the extent and duration of activation of kinases/substrates such as DNA-PK (T2609), ATM (S1981), Chk2 (T68), KAP1 (S824), SMC1 (S966) and p53 (S15) and by measuring specific phosphorylation sites. Activation of these

kinases/substrates will be quantified by western blotting and immunofluorescence staining with phospho-peptide specific antibodies. An altered pattern of these proteins after HZE particle radiation compared to non-irradiated controls would indicate early stages of carcinogenesis.

Project 4. Integrating Biomarkers for Mechanistic Understanding of Lung Cancer Pathogenesis and Risk Assessment after HZE Particle Irradiation. Yang Xie, Ph.D. (Project Lead), John D. Minna, M.D. (Project Co-Lead) Adi F. Gazdar, M.D. (Project Co-Lead).

The overall goal of our NSCOR is to provide the information required to develop a rational scientific basis for estimation of risks and mechanistic understanding for lung cancer occurring after exposure to radiation during space flight. This goal will be achieved by generating data from human lung epithelial cell and mouse models of radiation induced lung cancer pathogenesis and DNA damage (Projects 1-3). By integrating these findings using molecular analyses and functional studies, combined with systems biology approaches as well as our large legacy datasets, this project will generate a comprehensive model for lung cancer risk assessment after HZE particle irradiation. In addition, we will deliver a molecular toolkit for risk assessment of individual astronauts after space flights that will reduce the dependency on epidemiological data and population-based risk assessment and provide molecular tools of future aid to astronauts including means for early diagnosis and prevention of lung cancer.

Specific Aims:

Aim 1: Identify molecular signatures of HZE particle induced human and mouse lung epithelial cell transformation and compare these to molecular events of terrestrial developed lung cancers in never and ever smokers. Sub Aim 1.1: Systematically explore genome-wide genetic and epigenetic changes to identify genes associated with HZE particle induced human lung epithelial cell transformation to malignancy. Sub Aim 1.2: Identify genes and pathways that are unique to HZE associated lung cancer by systematically comparing genes associated with HZE particle induced lung cancer in human and mouse models and genes associated with terrestrial lung cancer.

Aim 2: Use computational biology and statistical approaches to identify “hub” and “master regulators” of tumorigenesis associated with HZE particle irradiation. This aim makes heavy use of the Database and biostatistical core of this NSCOR.

Aim 3: Develop and validate a risk assessment model for HZE associated lung cancer. Perform functional tests of selected genes through exogenous expression and siRNA, shRNA mediated knockdown approaches to identify genes whose changes in expression regulate the growth of HZE particle induced lung epithelial cell transformation.

Aim 4: Integrate findings in this NSCOR with those of the other NASA Lung Cancer Consortium members. Bayesian meta-analysis of gene set enrichment studies. Much attention has been drawn to the pathway or gene set analysis, which aims to detect altered biological pathways or other pre-defined gene classes rather than individual genes. Gene sets are usually defined based on gene functions, ontology information, chromosomal proximity, or known regulatory relationships. Gene sets that are altered in responding to changes in phenotypes or treatments may provide important insights into molecular functions and gene regulatory relationships in the underlying biological processes. Also, it has been reported that gene set analysis tends to yield more reproducible and interpretable results than single gene analysis. In order to identify pathways that are unique to HZE associated lung cancer, we developed a powerful computation tool for analyzing multiple gene expression datasets simultaneously.

It is challenging to integrate information of gene set enrichment analysis from multiple genomic studies targeting the same biological problem. Often the sample sizes of individual studies are not large and the microarray data are noisy, making estimation and inference highly variable. This often leads to inconsistent conclusions across studies. Integrative analysis of independent studies may greatly facilitate information sharing and improve the power of detecting truly enriched gene classes, as well as increase reproducibility and interpretability. However, direct combination of multiple microarray data sets such as stacking them into one is extremely difficult due to the incompatibility of data generated from various microarray platforms and different versions within the same platform. Therefore, meta-analysis, a systematic statistical synthesis of data from multiple studies, has been widely used to aggregate evidence from related studies. When data from component studies are available, using all data in a fully integrated, model-based framework is in general more efficient than a sequential approach using summary statistics only. Therefore, we developed a flexible Bayesian model that provide a natural way for data synthesis by incorporating model and parameter uncertainties involved in all studies. More importantly, it furnishes an integrated Bayesian framework for jointly modeling gene expression data from multiple studies and gene set information. This will allow researchers to conduct differential expression analysis, gene set enrichment analysis and meta-analysis simultaneously, all based on objective Bayesian posterior inference, which may yield more reliable scientific findings than the existing sequential approaches.

Integrate miRNA expression data to increase the prediction accuracy of radiation response in cancer patients. The goal of this analysis is to identify markers and signatures that add value to existing predictors of risk modeling. miRNAs, a class of post-transcriptional regulators, are short RNA sequences of 19-30 nucleotides, that bind 3’ UTRs of specific mRNAs. A single miRNA can target hundreds of mRNA to suppress the translation of mRNA by direct inhibition or by promoting degradation. There are also miRNA that activate genes. miRNAs are involved in the normal function of cells, and similar to mRNA, their deregulation is associated with disease. An increasing number of miRNAs have been implicated in cancer, having both oncogenic or tumor suppressor roles. As an example of how we plan to use miRNA expression signatures following HZE particle exposure, we developed a miRNA expression signature for cancer patients (including lung cancer and head and neck cancer patients) response to radiation treatment. The primary endpoint is recurrence free survival (RFS), which is analyzed using Cox proportional hazards models. Initial analysis using the Cox models on miRNA expression has identified candidate predictive markers. We are in the process of integrating the miRNA data with other clinical variables to predict radiation response in lung cancer.

Systems biology approaches to identify “hub” and “master regulators” of tumorigenesis associated with HZE particle irradiation. Gene regulatory networks (GRNs) describe the interaction among genes and how the genes work together to form modules of systematic understanding of molecular mechanisms underlying the biological processes. In the GRN, the genes that interact with many other genes are called “hub” genes. The “hub” genes that interact/regulate many transcriptional regulators are termed “master regulators”. The hub genes and master regulators are likely to be drivers of the disease status due to their key positions in the GRNs. As an example of how we are going to apply such a systems biology approach to look at the mRNA responses of lung epithelial cells to HZE particle radiation, we developed a novel supervised network approach that identifies both prognosis and predictive (to chemotherapy) signatures for lung cancer survival. We first identified 798 genes whose expressions are associated with patients’ survival time (controlling false

discovery rate at 10%) from 442 lung tumor samples in the lung cancer consortium dataset[99]. Then we constructed a gene network based on mRNA expression profiles of these 798 genes and identified 18 hub genes (genes with more than 6 connections). Interestingly, these 18 hub genes are significantly enriched (12 out of 18, p=0.005) with both functionally important “synthetic lethal” genes and genes with genetic alteration in lung cancer. Our study showed that these 18 genes contain more information (large variability in principal components) than other genes and can predict patients’ prognosis in an independent lung cancer patient’s cohort. We further validated the prognosis value of this 18-hub-gene signature in 7 independent datasets across different microarray platforms. More importantly, these genes can predict who will benefit from adjuvant chemotherapy (ACT) for early stage lung cancer patients: ACT significantly benefit patients in the predicted sensitive group, but not those in the predicted resistant group. In summary, through network approaches, we have identified 18 genes, which are highly (1) “functional”, (2) “informative” and (3) clinically important.

NSCOR Bioinformatics, Biostatistics and Database Core. (Y. Xie, Core Lead, J. Xie, Core Co-Lead; L. Girard Core Co-Lead).

The research proposed by the NSCOR encompasses a broad range of activities, including studies in cell lines and animal models utilizing many different types of radiation experiments. The Bioinformatics, Biostatistics and Database Core (Core B) has the expertise to effectively and critically analyze all of these types of data. This Core assists Project investigators with hypothesis refinement, experimental design, database development and management, quality control, analysis and informative presentation of results across all projects of the NSCOR. Members of the Core participate in monthly NSCOR meetings to ensure that proper consideration of biostatistics and data management are considered prospectively during all phases of NSCOR experiments.

Specific Aims:

Aim 1: To provide valid statistical designs of experiments and conduct innovative statistical modeling, simulations, and data analyses.

Aim 2: Data integration and quality control.

Aim 3: To expand the current UTSW-NSCOR database to an ontology-driven and internet accessible database to allow data sharing and integrative analyses.

As we progress on all the NSCOR projects, significant amount of new data have been generated since our 2010 annual progress report. As a result of Core B effort, a comprehensive data management system has been established that allows efficient access and sharing of all NSCOR data.

Experiment Planning and Data Analysis: NSCOR Core B: Biostatistics and Bioinformatics is working on the following in terms of experiment planning and data analyses: Formulation of testable hypotheses based on data already generated in the various Projects.

On Project 1: HZE Particle Exposure and the Risk for Human Lung Carcinogenesis. Using novel human cellular models for lung cancer we have developed, this project will: a) Identify the radiation parameters that drive radiation-induced cellular transformation to malignancy in normal respiratory cells from the central and peripheral airways; b) Determine the role of gender, lung cell of origin, and inter-individual variation in HZE particle induced lung carcinogenesis; and c) Determine the nature and frequency of genetic changes, and pulmonary stem cell markers associated with HZE particle-induced lung cancer. All of these results and unique bioreagents will be made available to other NASA Lung Cancer Consortium members. Genetic and epigenetic changes in HBE following HZE irradiation – the genome wide studies of DNA copy number and methylation changes are just beginning and we are helping with the statistical plan for these experiments. The Illumina beadarray microarray data was summarized using Illumina Beadstudio software and then processed using the Model-based Background Correction for Beadarray (MBCB) method, which was developed by the biostatistician of the Core D and the investigators of Project 1 [1, 2, 7]. The quantile-quantile normalization [3] was then performed on the log transformed gene expression data. Hierarchical clustering and histogram were used for quality control. The differentially expressed genes were identified using Samr package (Significance Analysis of Microarray) [4] in R statistical software [5]. The multiple testing issues were addressed by implementing q-values to estimate the gene specific false discovery rate (FDR) [6]. In this study, the genes with FDR less than 0.05 were considered as differentially expressed. Both multi-dimensional scaling (MDS) approach and principle component analysis (PCA) approach were used for dimension reduction. Ingenuity Pathways Analysis tools (Ingenuity system Inc.) were used to identify the pathways that were perturbed after radiation treatments.

On Project 2: Mouse Models of Lung Cancer after HZE Particle Irradiation. Using transgenic mouse models of lung cancer this project will: a) Determine the role of inflammatory cell responses in murine lung cancer arising after HZE particle exposure; b) Using genome wide technology, compare the genetic changes in radiation induced vs. spontaneous arising murine lung cancers. All of these results and unique bioreagents will be made available to other NASA Lung Cancer Consortium Members. In planning the animal experiments on evaluating the RBEs for different dose regimens of X-ray and HZE particles , we based on survival rates at a pre-specified time point. Pre-determined margin of error was used to calculate the necessary numbers of mice for each cohort. For low- and high-LET irradiated mice, we have estimated the empirical distributions of the survival time using the Kaplan-Meier method and compared these survival curves among experimental conditions and cohort groups using the non-parametric Log-Rank test. When fitting multivariable survival regression models, Cox proportional hazard model is chosen over the parametric regression model due to the often encountered difficulty of selecting an appropriate parametric hazard distribution. The proportional hazards assumption of Cox model is examined graphically and regression diagnostics of the model (residual analysis) is conducted to ensure the model’s appropriateness. Number of tumor lesions and frequencies of different tumor types among different irradiation regiments are compared using Mantel-Haenszel test, chi-square test or Fisher’s exact test. Tumor lesion incidence rates are adjusted by survival time when needed. We helped analyzed and interpreted both mouse survival data and genomics data, and co-authored a paper.

On Project 3: Organotypic (3D) Human Models of Genomic Instability for HZE Particle Induced Lung Carcinogenesis. This project utilizes novel human cell based models to: a) Demonstrate that the HZE particle induced persistent DNA damage and genomic instability (by monitoring of 53BP1 foci) in human organotypic lung respiratory cell models is dependent on the dose, dose rate, energy, “Z” number and frequency (acute versus fractionated) of HZE particle radiation; b) Determine the mechanisms of HZE particles induced human lung carcinogenesis in the organotypic model. The results and novel bioreagents created by this project will be shared with other NASA Lung Cancer Consortium

members. We participated in the experiment design and data analysis for this project. Core B is participating in these analyses and is working with Project 3 investigators in formulating specific testable hypotheses.

On Project 4: Integrating Biomarkers for Mechanistic Understanding of Lung Cancer Pathogenesis and Risk Assessment after HZE Particle Irradiation. This project integrates data from the other three projects as well as data generated from other unique and relevant non NSCOR funded sources ("legacy databases"), identifies key pathways, "hub" and "master regulator" genes, validates their biological importance and develops a risk model for space radiation induced lung cancer. Its aims are: a) Identify molecular signatures of HZE particle induced human and mouse lung epithelial cell transformation and compare these to molecular events of terrestrial developed lung cancers in never and ever smokers; b) Use of computational biology and statistical approaches to identify "hub" and "master regulators" of lung tumorigenesis associated with HZE particle irradiation in human cells and murine models of lung cancer. All of the results and novel bioreagents from this project will be shared with other NASA Lung Cancer Consortium members. Core B is working closely with investigators on Project 4 to develop bioinformatics tools and systems biology approaches for data integration. We have developed ProbeMapper database for data integration and systematically compared statistical methods for constructing gene networks and identifying hub genes. ProbeMapper, developed by investigators in this core, is to do data annotation and alignment across different microarray platforms. The molecular profiling data from different platforms will be aligned using a BLAST alignment of the probe sequences, and the most recent revision of the Transcriptome will be used to give an accurate annotation for each probe on each platform.

Developing novel computational approaches to construct gene network and identify hub genes. During the last NSCOR research period, we compared six popular computational methods to construct GRNs from gene expression data. Our results show that these algorithms performed reasonably well especially when the number of microarray becomes large. However, no method uniformly outperforms other approaches. Since each method has its own advantages, we propose to use a joint analysis method to aggregate the results from different methods. In addition, we have developed an ensemble-based network aggregation (ENA) method that can further improve the performance by aggregating the networks built from the bootstrap samples. Based on these results, we will use the ENA method to construct a lung-cancer-specific gene network based on different computation methods and different data sources. For data sets 1,…,M (M is the number of datasets), and for approaches 1, …,N (N is the number of computational approaches), we will score the connectivity between gene X and gene Y separately. Then we will use a non-parametric method to aggregate the results from different methods and datasets as the final connectivity for each gene pair. The gene-gene interactions in the final GRN will be determined by their respective connectivity scores. The hub genes will be identified based on this gene network. We will apply this new approach to identify hub genes associated with HZE induced lung cancer.

On Exploratory Data Analysis. For a large interdisciplinary joint research effort such as our NSCOR investigation, exploratory data analysis is not only unavoidable but also indispensable. We rely on our pre-specified primary hypotheses for confirmative analyses. But due to the wealth of data that are being generated from the UTSW NSCOR multi-projects, enormous new information will be obtained to allow us, on one hand to possibly refine our existing models and on the other hand, to elucidate new relationships among different observations. The Biostatistics and Bioinformatics Core is working closely with all of our project investigators to fully take advantage of the large body of new data, yet to guard against drawing definitive conclusions from this hypothesis-generating type of analysis. Explorations on relationships among new variables will also help us in eliminating possible multi-colinearity problems when regression analyses are conducted.

On Data Integration. We have developed a method [78] to model genome-wide DNA binding, mRNA expression and sequence motif data in a study to identify the binding targets of genes. Both simulation studies and an analysis of an experimental data set show that the joint modeling method can significantly improve the specificity and sensitivity of identifying target genes as compared with conventional approaches relying on a single data source.

Purpose and Description of the NSCOR Database. The NSCOR experiments are generating vast amounts of information in the form of computer data files (raw and processed data, analyses, images) and biological material. It is crucial to organize this information in a form that allows efficient and secure data storage and sharing. Additionally, it is important to keep track of all the biological samples (e.g. vials of cells collected at various time points following irradiation) so that errors are diminished and experiments are performed efficiently. Finally, a complete repository of all NSCOR data constitutes one of the deliverables of this project. We have therefore constructed a database for NSCOR that has a server-client architecture and that can store and share data generated from the various NSCOR experiments. It has three components: a) a server (built on a quad-core computer with 8-GB RAM and 3-TB hard drives) that is accessible to all NSCOR investigators. It has folders for each projects and cores to store and organize the experimental data and related or administrative information, b) a Microsoft Access database also located on the server and containing tabular descriptions of experiments as well as lists of samples (DNA, RNA, cell vials, etc.) with associated information including physical storage location, c) various Excel clients located on each investigator's computer that serve as the primary interface between the users and the central (Access) database. The Excel clients contain data query and editing functionality and each change to the database is immediately transferred to the central database, and from there to the other clients so that data remain synchronized at all times. A log table of all changes made to the database is also included and can be used to keep track of samples and experiments, as well as maintaining data integrity and facilitating communication between investigators.

Database Access. The server is accessible as a network drive (PC and Mac), facilitating data exchange between server and local computers. Remote connection is also possible through VPN (Virtual Private Network) which preserves the same interface together with all database functionalities. This is especially useful when performing radiation experiments at a remote location (such as Brookhaven National Laboratory) and numerous sample data need to be entered in the database.

Data Collection. More than 17,000 samples (including cell pellets and cryogenically preserved cell lines) and experiments (irradiation, microarrays, mouse data) have so far been collected and stored in the NSCOR database. These data can be viewed or edited by any participating investigator.

Data Sharing and Pre-processing. Having all data in one place makes it possible to develop consistent approaches for data analysis and processing. As an example, we implemented a tool within the NSCOR database where microarrays can be selected by a user, array pre-processing and normalization can be performed, and the selected arrays displayed in a spreadsheet, which can then be analyzed further with any third-party application. The advantage of this functionality is manifest in increased efficiency, both in the selection of arrays and their pre-processing, as well as in consistency of analysis between collaborators.

Administrator-level Functionality. Database management functionality is being developed so that one or more additional administrators can participate in managing the database without the need for programming experience or Microsoft Access knowledge. Specifically, some commands accessible only by administrators will be used i) for managing usernames and passwords, as well as their permission levels, ii) for backing up and restoring the complete database, iii) for modifying table structures such as adding new tables, creating or changing fields, iv) for installing a server-client database de novo with new data, as would be required for other NSCORs wishing to establish their own database using this system.

Database Summary. The database allows easy access and sharing of all NSCOR data by investigators. Several thousands of collected samples and large sets of data can be organized and shared by investigators and statisticians. Its centralized nature makes it possible to schedule regular backups, to ensure that data are organized in a way that analysis and data interpretation are facilitated, and to allow investigators to keep track of experiments and samples and thereby minimize the chance of errors. The database is accessible through the Internet making it possible to enter or edit data in real-time from remote locations. Finally, the database can be disseminated, in whole or in part, to outside collaborators or reviewers and thereby facilitate data integration among various NSCOR projects.

Database Update. We are expanding the capability of the current UTSW-NSCOR server-client database into an organized, relational database repository for all datasets collected and processed by the projects and cores, which will serve as a public platform for data integration, analysis, sharing and management. The goal is to capture raw and processed data (both private and public) and organize and integrate them for easy access, sharing and utilization by NASA investigators. This database, associated tools, and data mining techniques, will be available seamlessly to all NSCOR investigators and will facilitate intra- and inter-NSCOR collaborations as well as collaboration with individually funded NASA investigators.

Core A: Administrative. Core Leads: John Minna and Jerry Shay. UTSW NSCOR Administrator: Brenda Zielke.

The Administrative Core has established accounts and control over disbursement of all funds from this NSCOR application. It has also kept track of all personnel employed by this NSCOR. It has provided a central place for communication and coordinated data sharing In addition, it has established a series of frequent scientific planning meeting of the NSCOR investigators at UTSW. Approximately 20 scientific personnel attend each of these meetings. At each meeting administrative matters are discussed, experiments are planned, administrative changes made by the senior leadership team, and most importantly a scientific presentation by one of the NSCOR investigators is made and critiqued by the other investigators. In addition, the Administrative Core worked with other lung cancer NSCORs such as Duke and Emory to have interactive meetings. One of these was hosted by UTSW and the other by Duke. In addition, Drs. Minna and Shay worked with NASA and the NCI to help host a meeting at the NCI for risk of developing lung cancer after CT scan or HZE particle (space) radiation. Dr. Minna serves as the Administrative Core Lead with Dr. Jerry Shay as the Administrative Core Co-Lead. The Administrative Core coordinates all of the arrangements for the participation of several NSCOR participants for travel the NASA investigators meeting in North Carolina in 2012. The Administrative core also arranges Internal Scientific Advisory Board Reviews and External Advisory Board Reviews. Finally, the Administrative Core coordinates travel of multiple personnel to BNL for experimental work.

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